

Histology of skin: hair, nails, glands and sensory receptors

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Do.
Make.
Heal.

Innovate.
Reinvent the Future.

Session Objectives

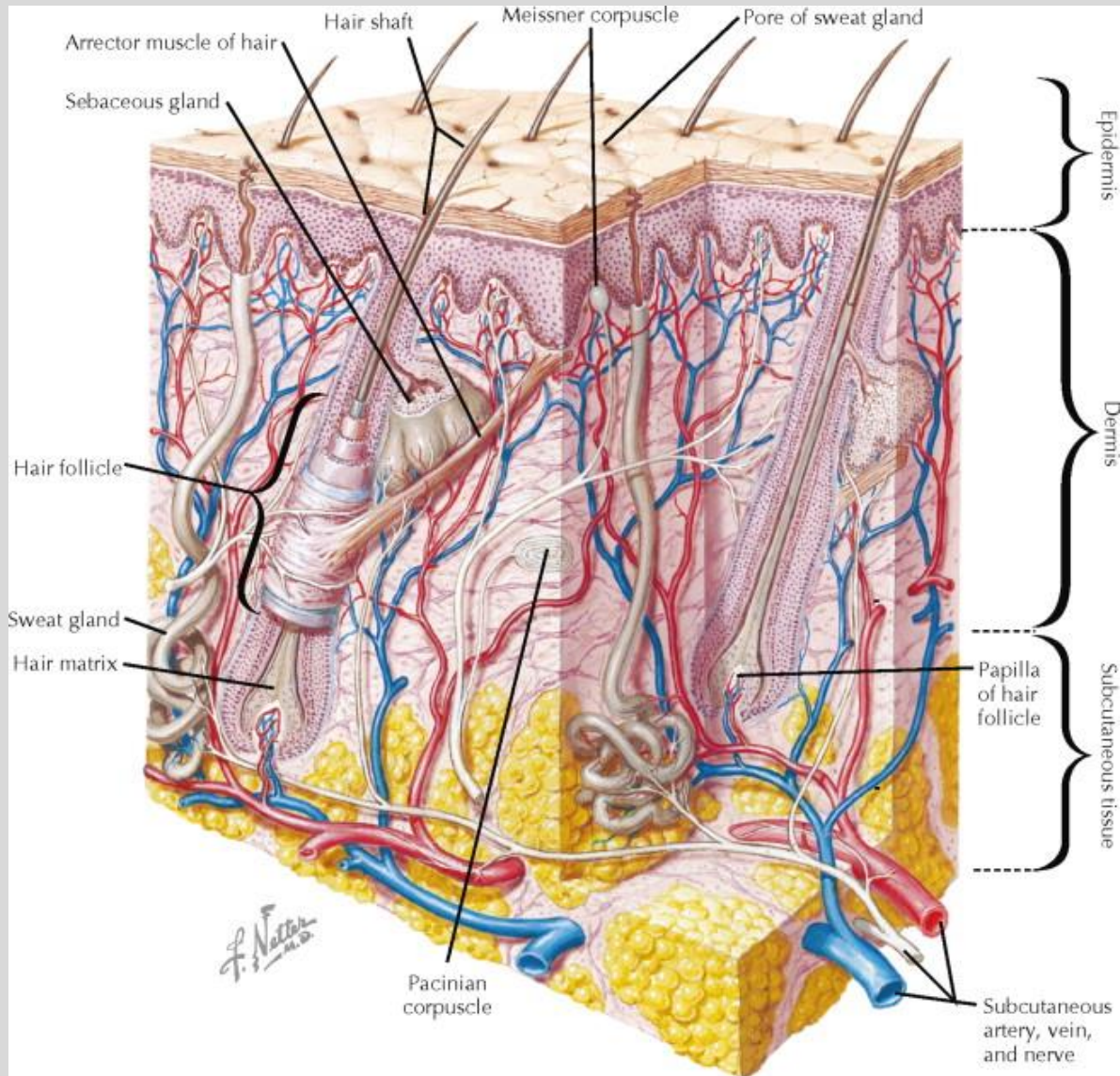
1. Contrast the major divisions of the hair complex and its relation to neighboring tissue.
2. Explain the development and life cycle of the hair complex.
3. Contrast the major divisions of the nail complex and their relation to neighboring tissue.
4. Explain the development of nails.
5. Describe the regional distribution of sweat and sebaceous glands, contrast their functions, and relate these functions to the cellular composition.
6. Locate and describe the variety of sensory receptors throughout the skin.



Appendages of the skin

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▲ Schematic of skin and its appendages that shows the epidermis, dermis, and subcutaneous tissue.

Source: “Integument” in Netter’s Essential Histology (2013)

Epidermis is derived from ectoderm during embryonic development. Cells of ectoderm (forming epidermis) grow downward into the dermis and give rise to:

- Hair follicles
- Fingernails
- Toenails
- Sweat glands
- Sebaceous glands
- Mammary glands



Hence, “epidermal appendages”.

Planting the seeds (hair)

1. Develop in 3rd month of fetal life.
2. Epidermis sends down-growth into dermis.
3. Down-growth becomes hair follicle. Invading epidermis forms a cap over the connective tissue papilla.
4. As development proceeds a mature adult follicle is formed and can continuously grow via keratinization.

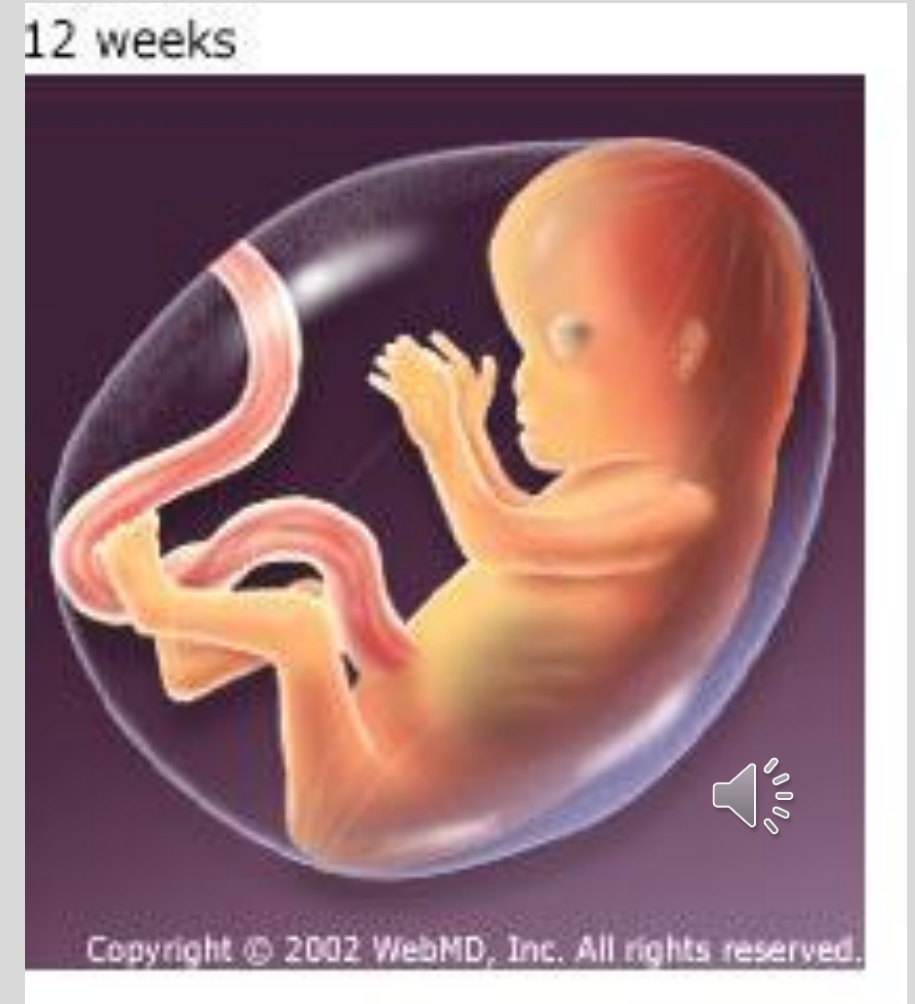
*similar developmental program as epidemis



Source: WebMD

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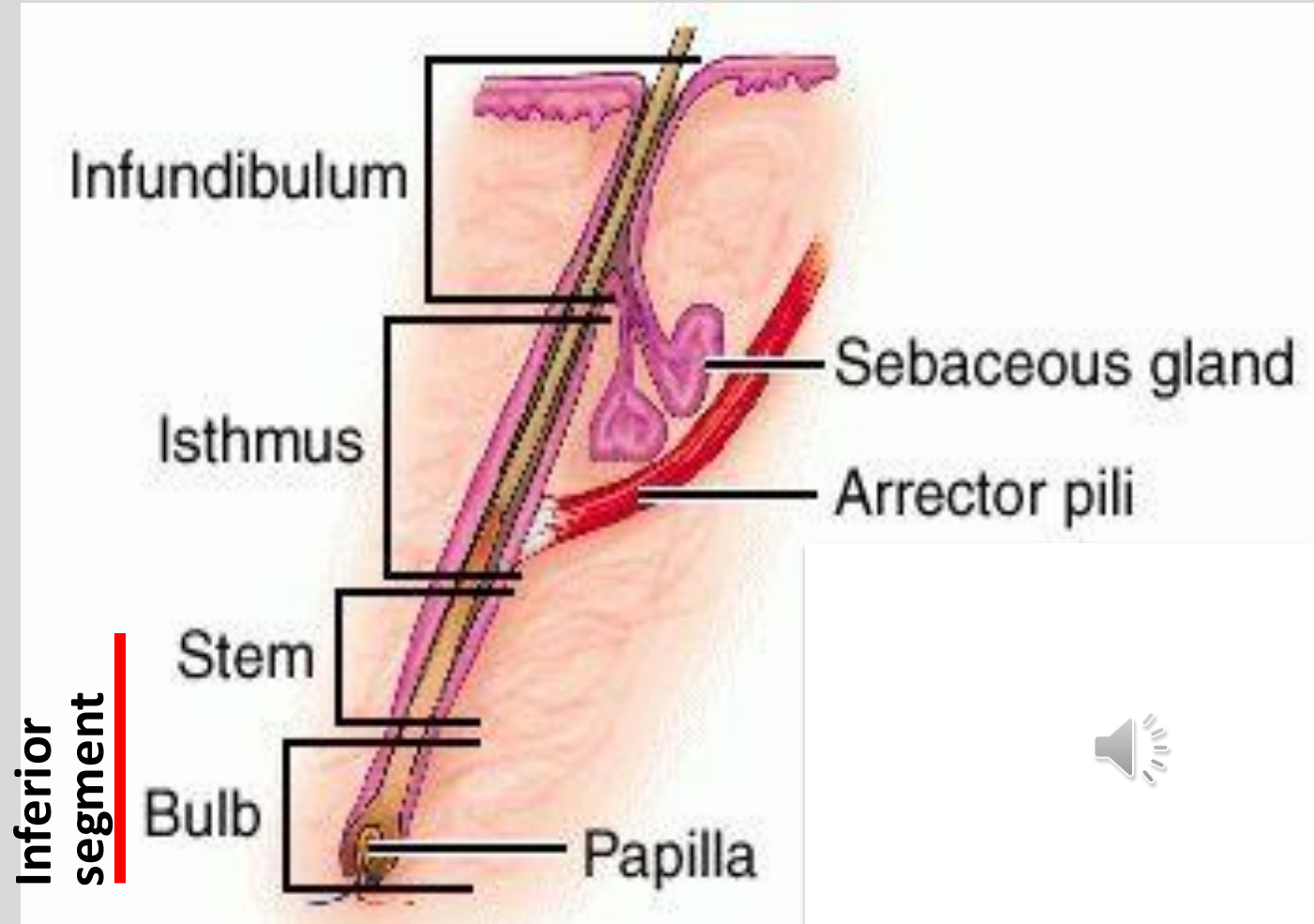


Three major segments of the hair follicle

Infundibulum - from surface opening of follicle to opening of its sebaceous gland

Isthmus - from infundibulum to level of insertion of the arrector pili muscle.

Inferior segment - growing follicle portion; at base expands to bulb invaginated by vascularized loose connective tissue called dermal papilla.



The Pilosebaceous Unit

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All mammals have sebaceous glands



no

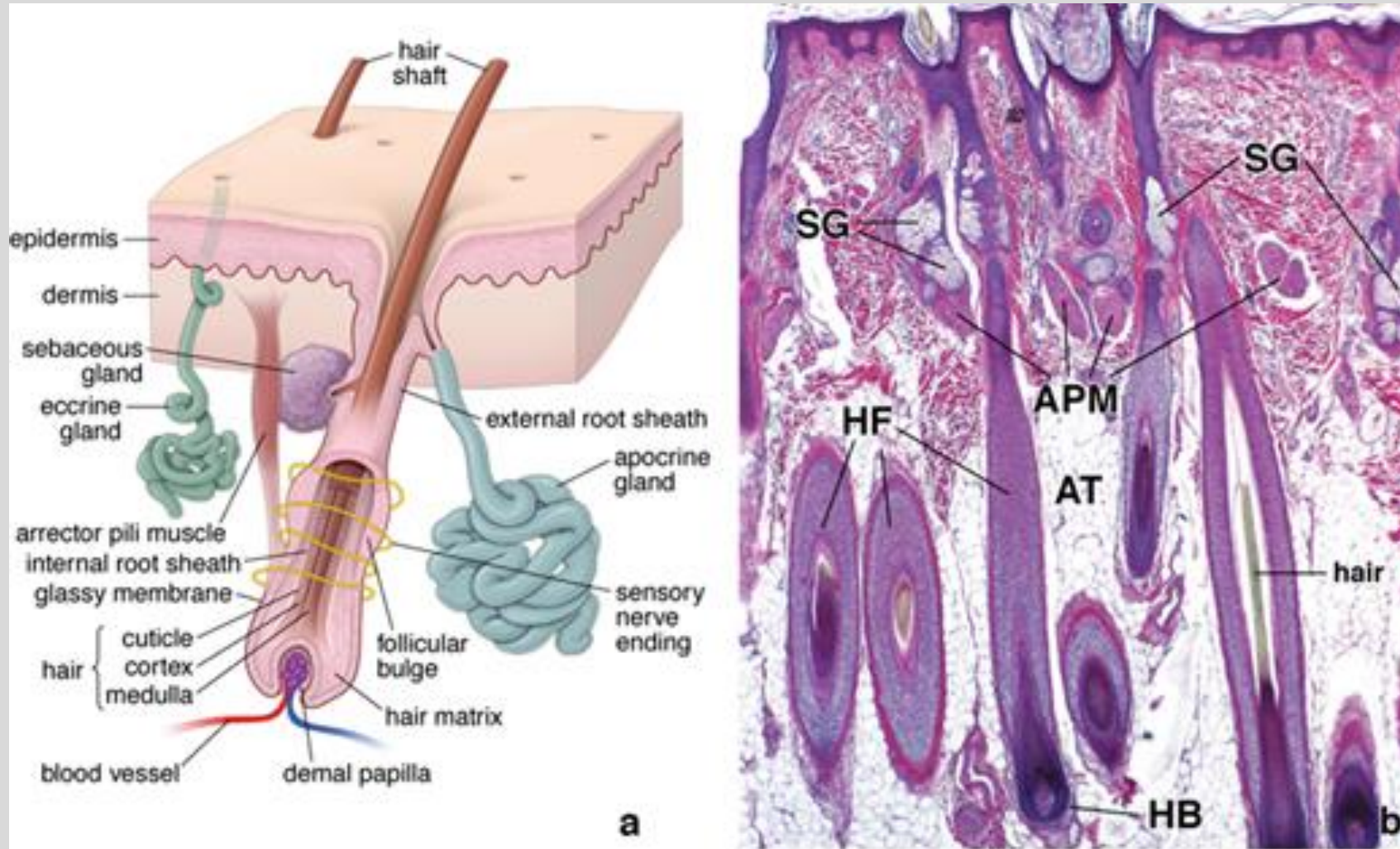


yes

The Pilosebaceous Unit, a “mini-organ”

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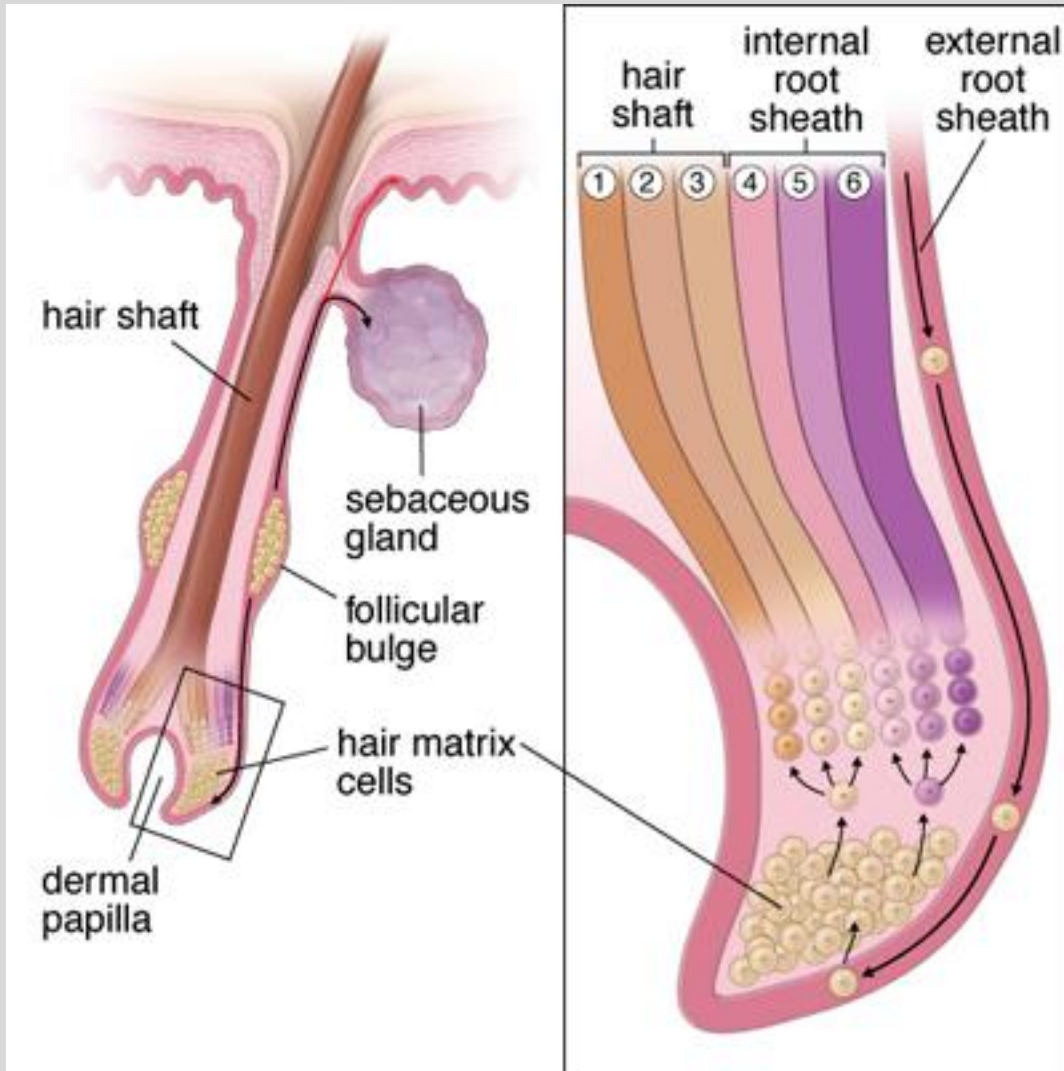


Schneider et al (2009)  The hair follicle as a dynamic miniorgan. Curr. Biol.

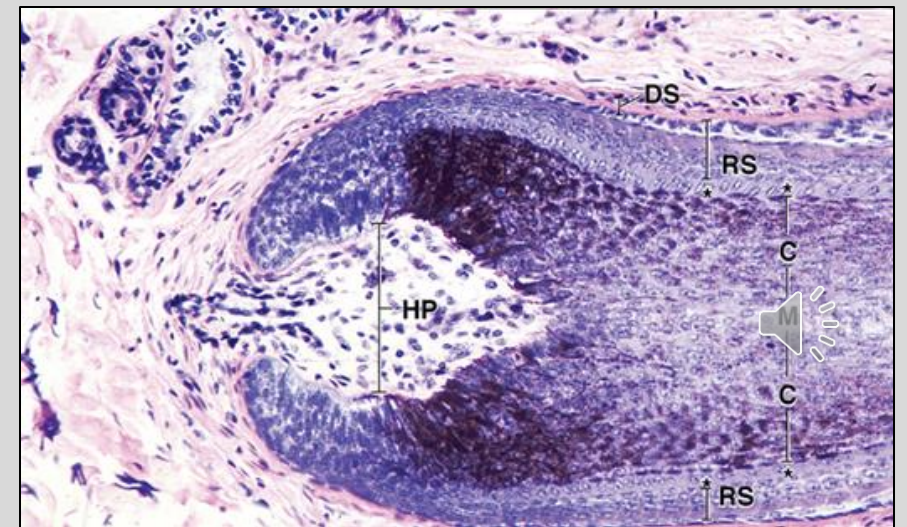
Follicular development

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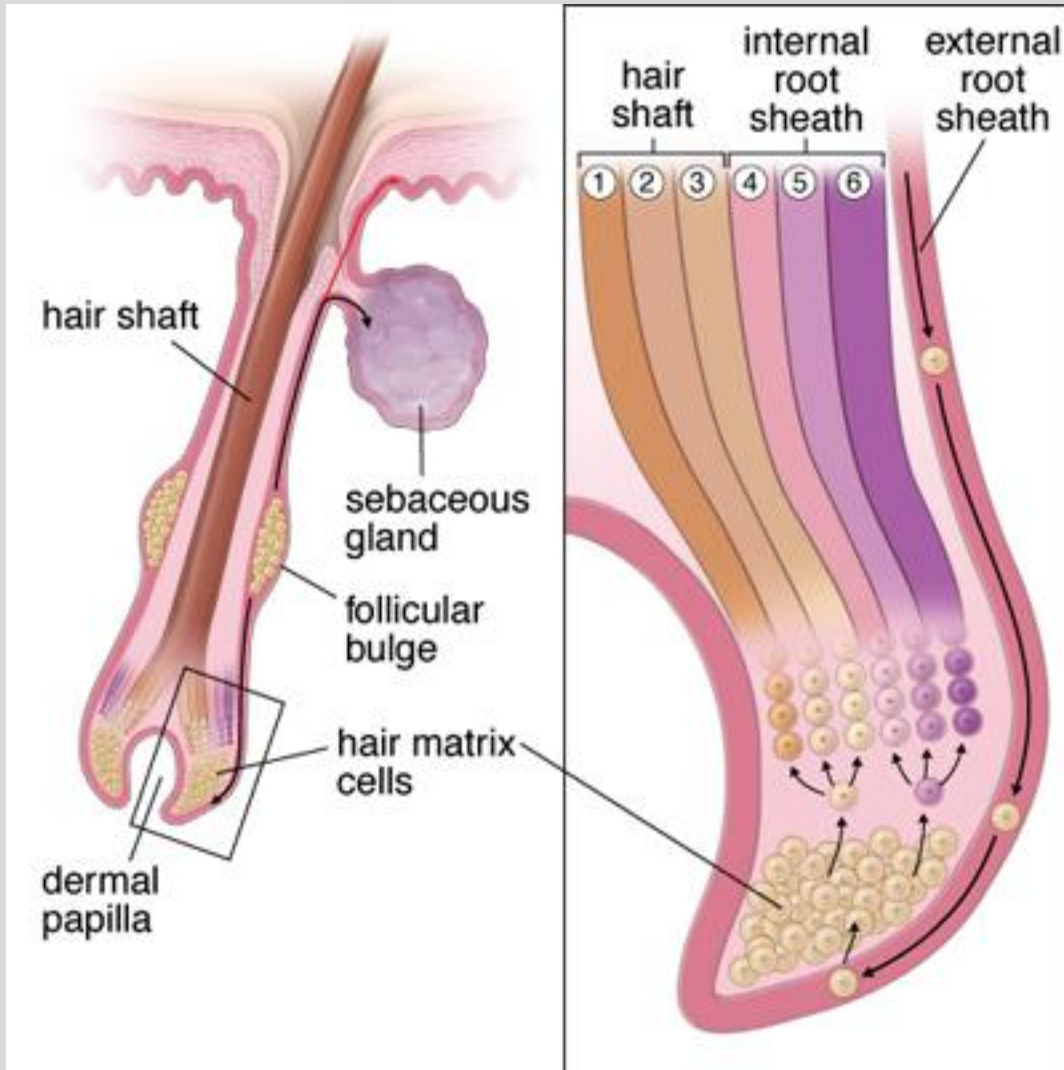
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- Follicular bulge houses epidermal stem cells which give rise to sebaceous gland and the hair follicle.
- Deepest downgrowth becomes germinal matrix of hair, includes keratinocytes and melanocytes
- Gives rise to the hair shaft and internal root sheath

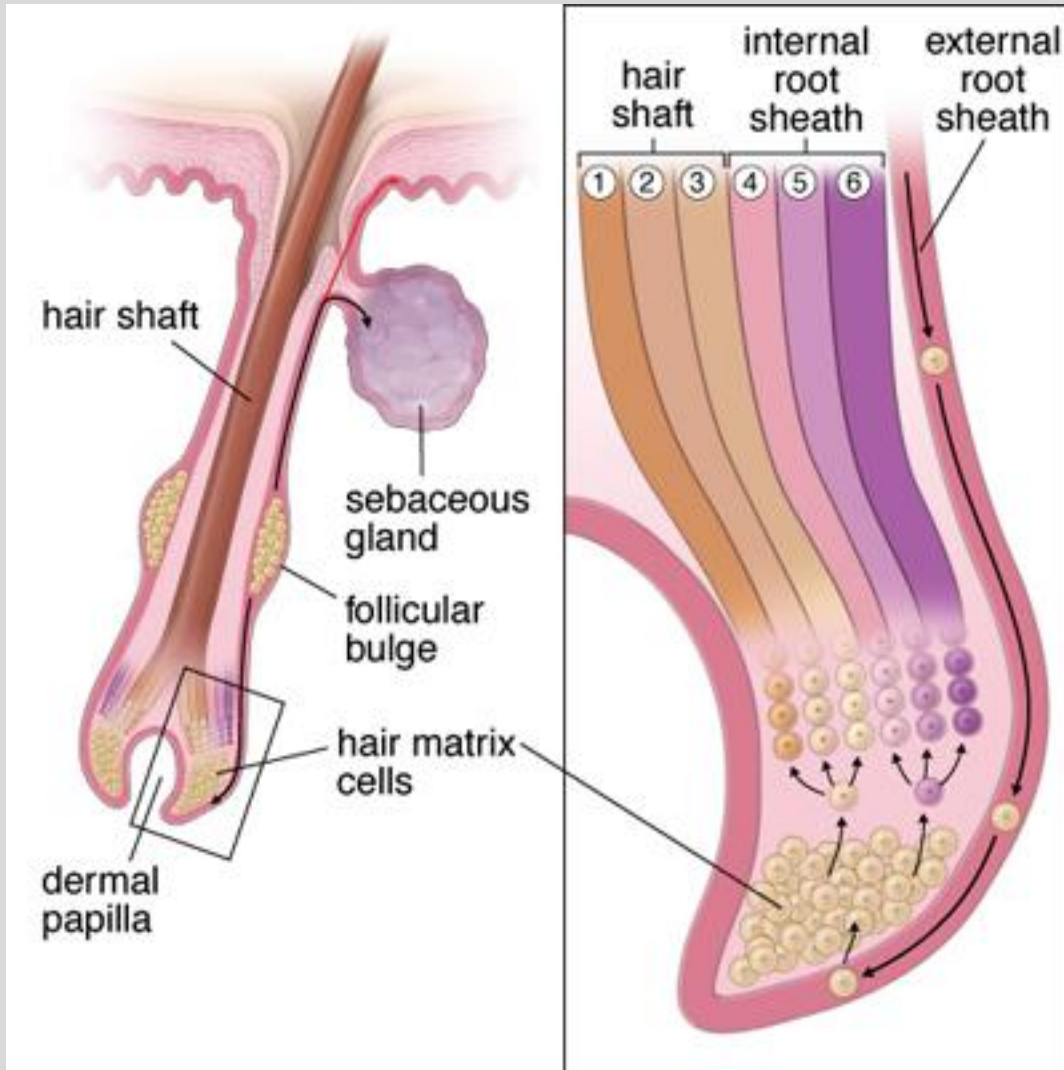


Follicular development



- Keratinization occurs after cells leave germinal matrix in region known as keratogenous zone
- When hair emerges from follicle it is entirely keratinized as hard keratin (analogous to entering stratum corneum), does not desquamate.
- Internal root sheath is soft keratin, is broken down/desquamates at opening of sebaceous gland

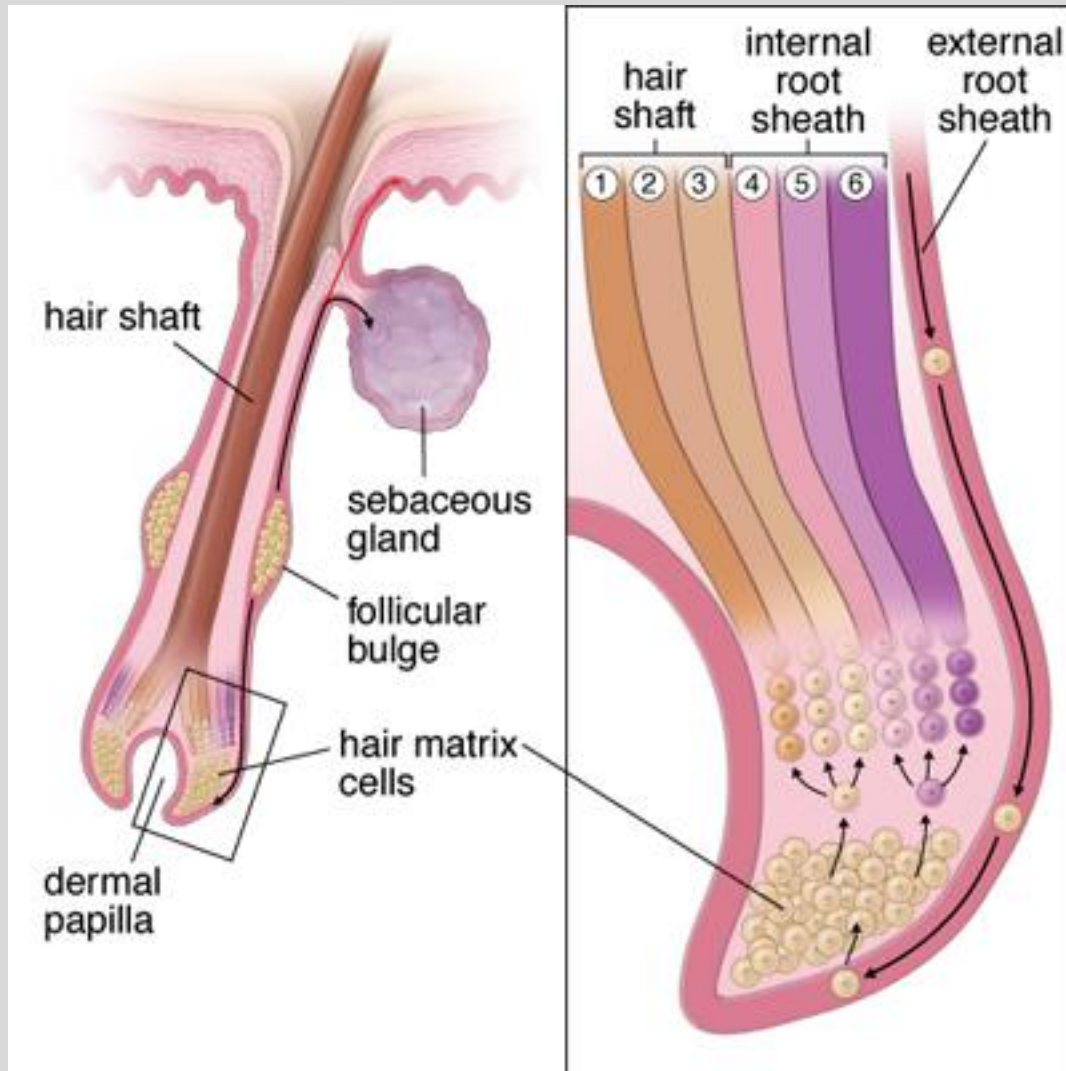
Multiple layers of the hair shaft



1. Medulla – central, soft keratin, large vacuolated cells; only present in thick/terminal hair
2. Cortex – peripheral to medulla, cells differentiate into keratin-filled cells; contains pigment, 80% mass
3. Cuticle of hair shaft – outermost layer of hair composed of very thin scale-like cells



Multiple layers of the internal root sheath



Need not remember these divisions

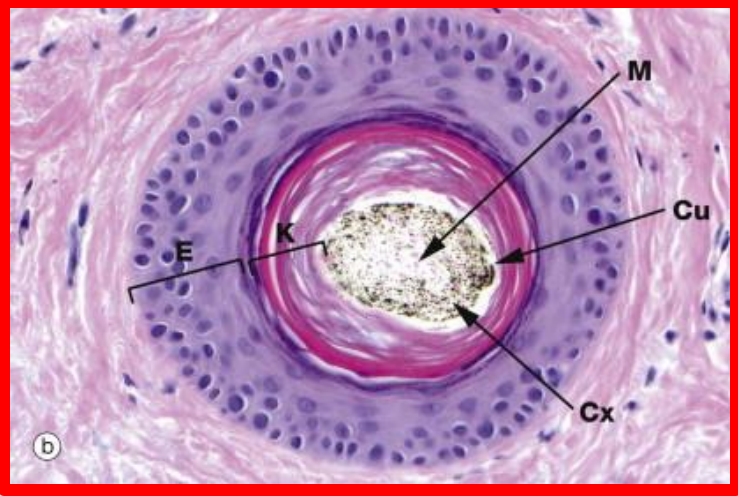
4. IRS cuticle: squamous cells

5. Huxley's layer: flattened cells

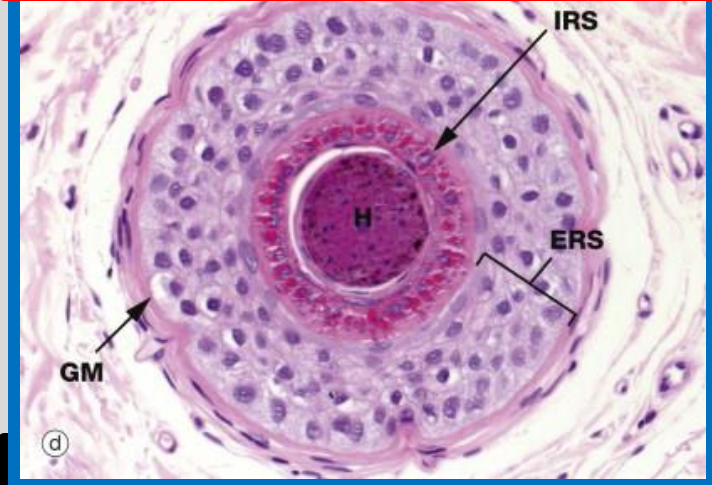
6. Henley's layer: cuboidal cells in direct contact with outermost part of hair follicle, the external root sheath

The internal root sheath is soft keratin and is broken down at infundibulum level where sebaceous secretions enter follicle

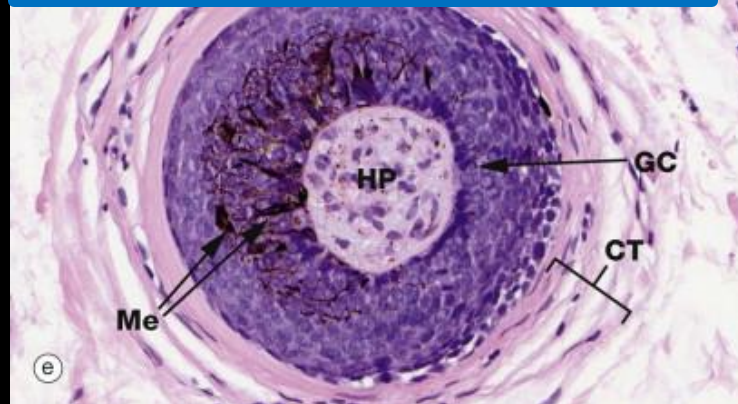
The external root sheath is epidermis and its basement membrane is called the glassy membrane



M - medulla
Cx - cortex
Cu - cuticle
E - epidermis



H - hair
IRS - internal root sheath
ERS - external root sheath
GM - glassy membrane



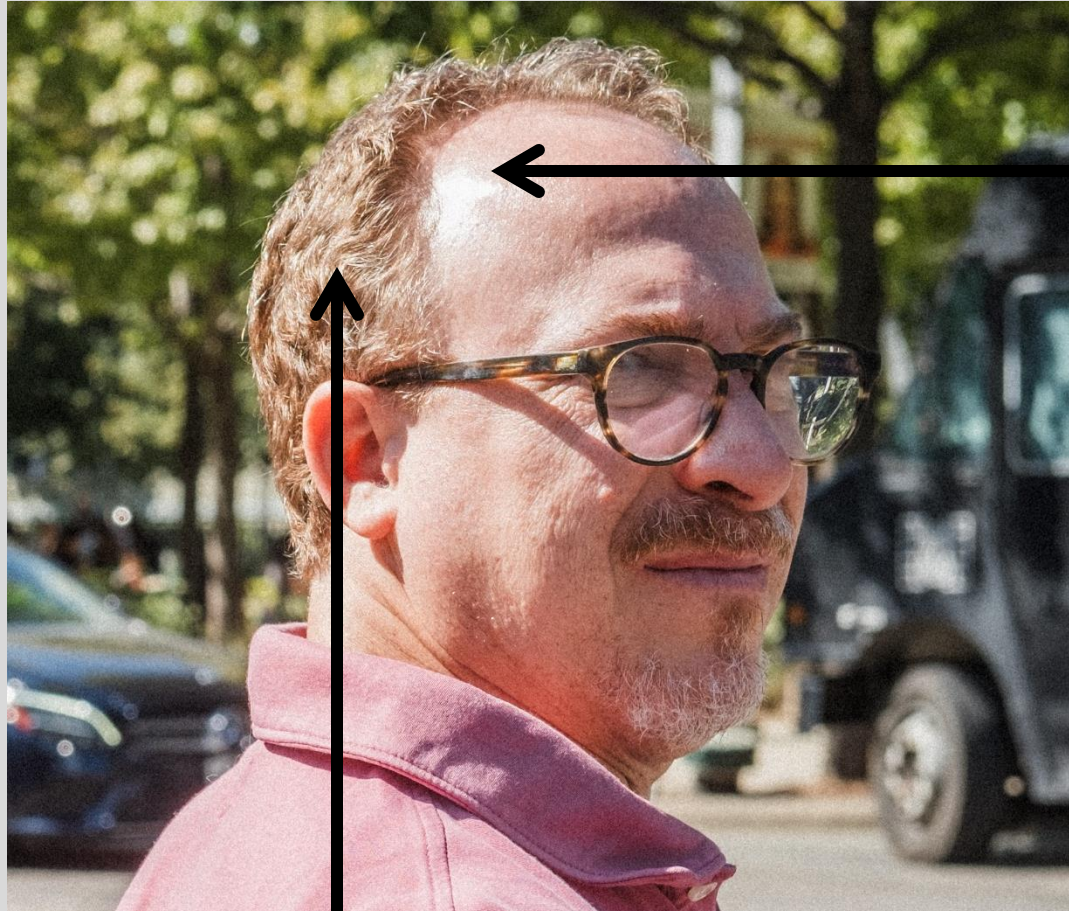
HP - hair papilla
GC - germinal cells
Me - melanocytes
GM - connective tissues



Vellus versus terminal hair

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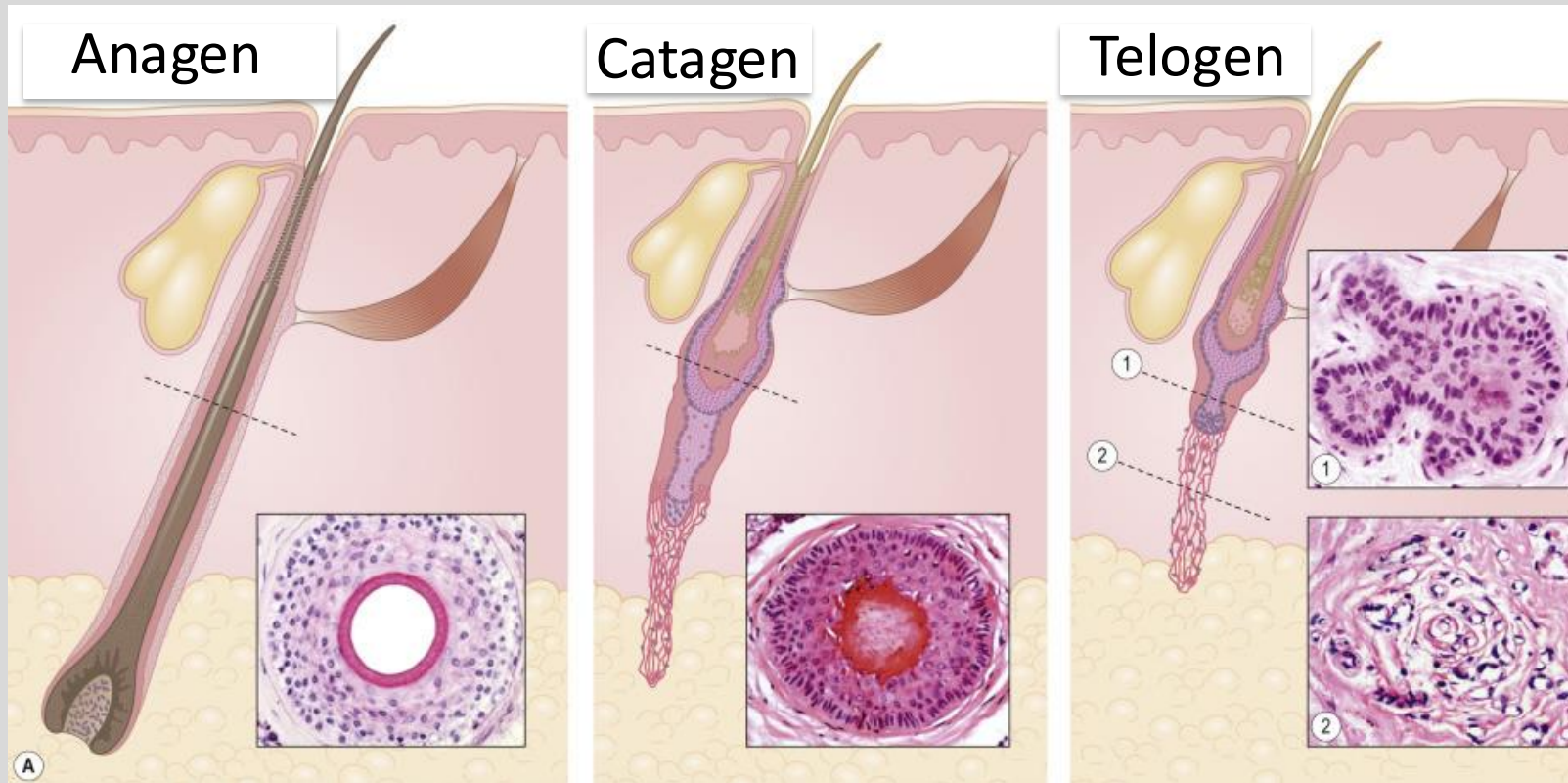
Vellus - thin fiber, follicle housed in dermis, distributed over the body except portion of external genitals, palms, and soles

Terminal - thick fiber, follicle housed in hypodermis, distributed to head, eyebrows, end of eyelids, genitals, axilla and other regions impacted by androgens during puberty

Phases of hair growth

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Below shows phase durations for hair on the scalp.
Course body hair and pubic hair has short anagen phase and long telogen phase

- Anagen (active growth period): 2-7 years, continuous growth, abundant melanin (if any).
~85% of hair follicles at any time
- Catagen (involution period): massive apoptosis, 2-3 weeks. 1-2% of follicles
- Telogen (rest period): final involution, ~100 days. ~15% of follicles

Arrector Pili (muscles for hair)

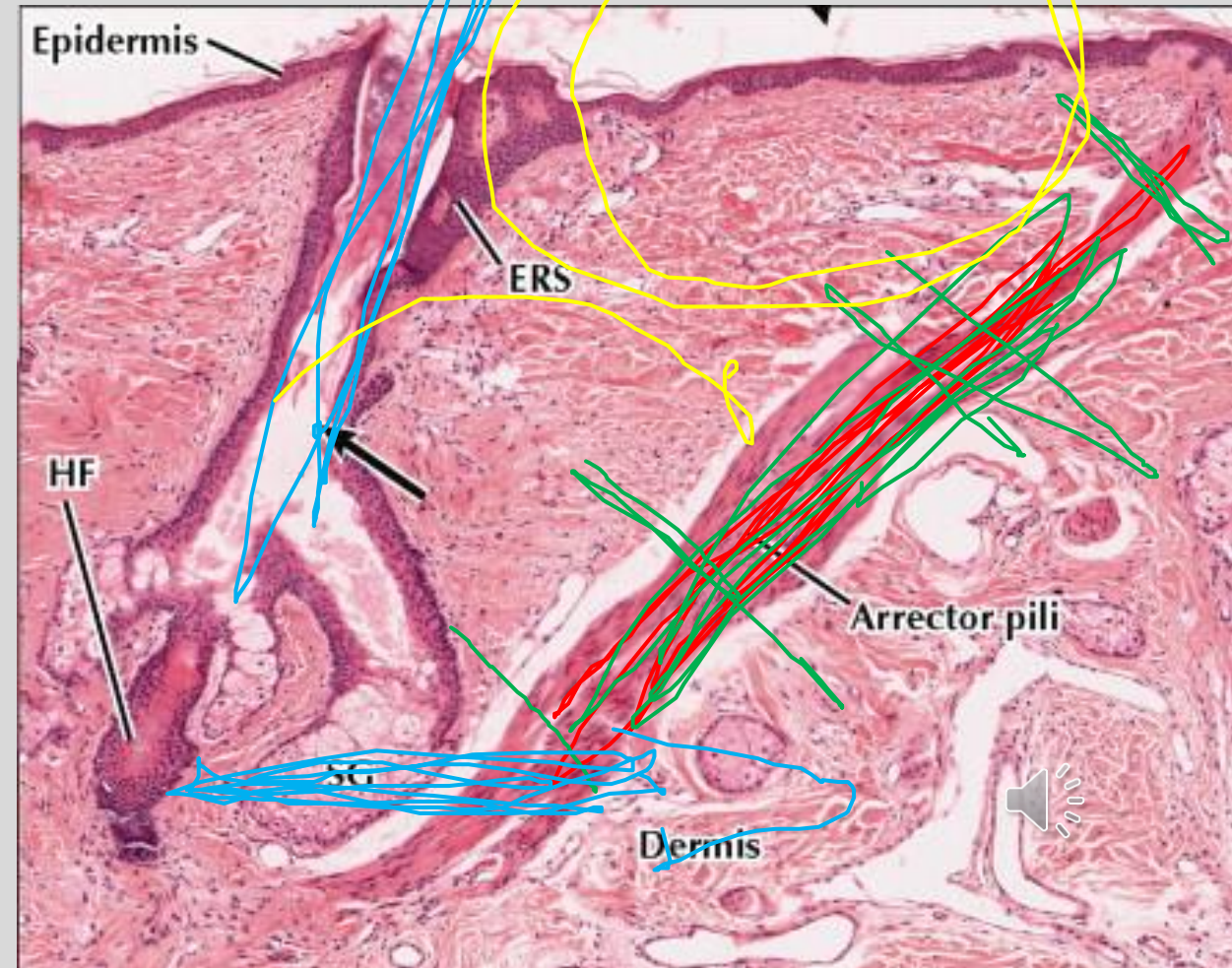
Bundle of smooth muscle fibers that attaches to the connective tissue sheath of a hair follicle

When muscle contracts, pulls hair follicle outwards and makes hair stand up; hairs exit at an angle.

Produces “goose bumps”/dimpling of skin

Also squeezes on sebaceous glands causes the secretions to be emptied

Activated by autonomic nervous system.



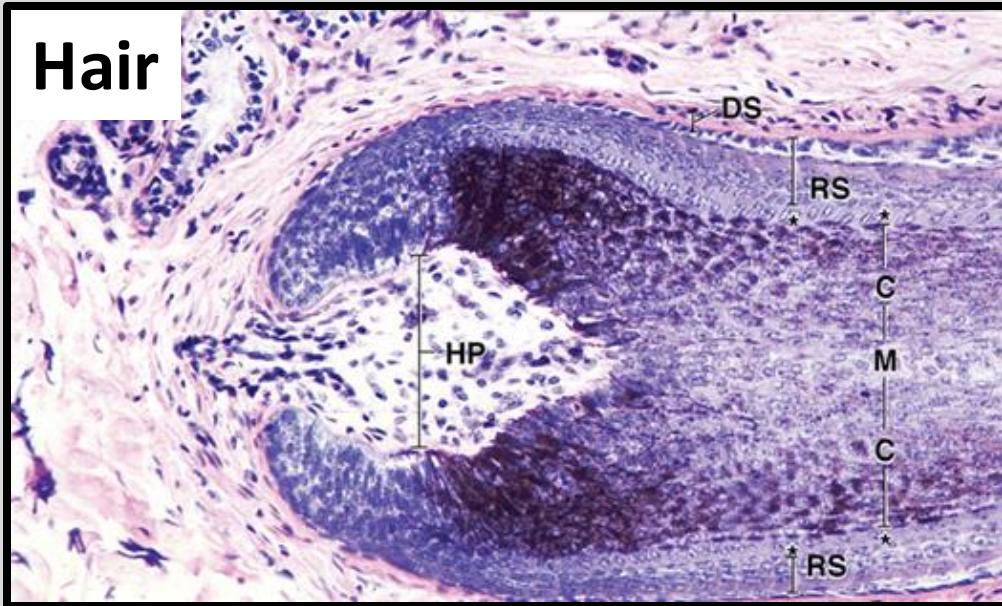
Source: “Integument” in Netter’s Essential Histology (2013)

Planting the seeds (Nails)

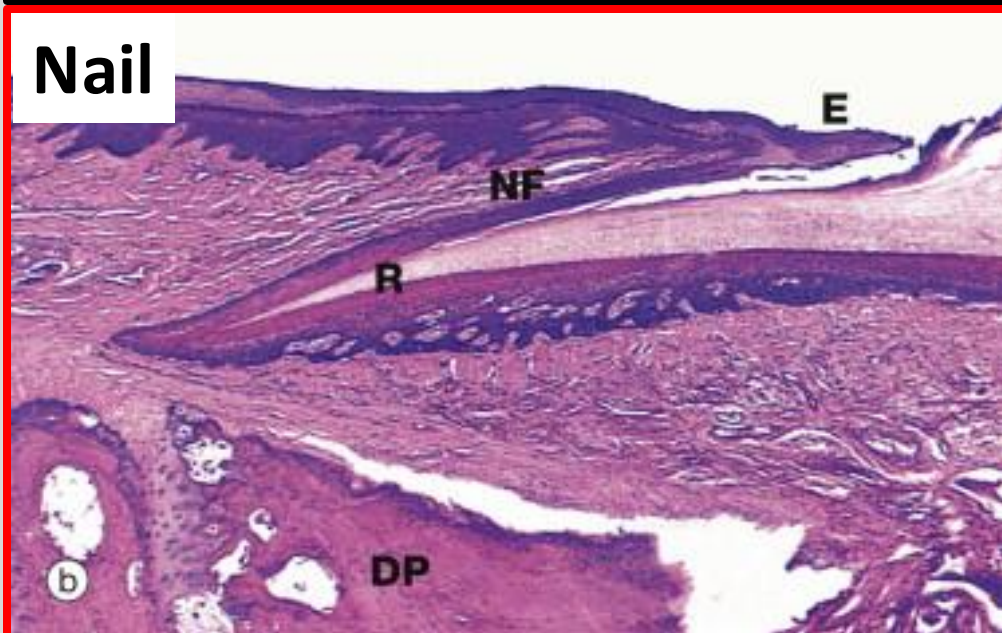
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Hair



Nail



3rd month of embryonic life, epidermis of fingers and toes “invade” underlying dermis along a curved line to form the nail

Proximal part of nail, the nail root, is buried in fold of epidermis covering cells of the germinative zone.

Cells become fully keratinized.

Source: “Integumentary System”
in Ross and Pawlina
Histology, 7th Ed. (2015)

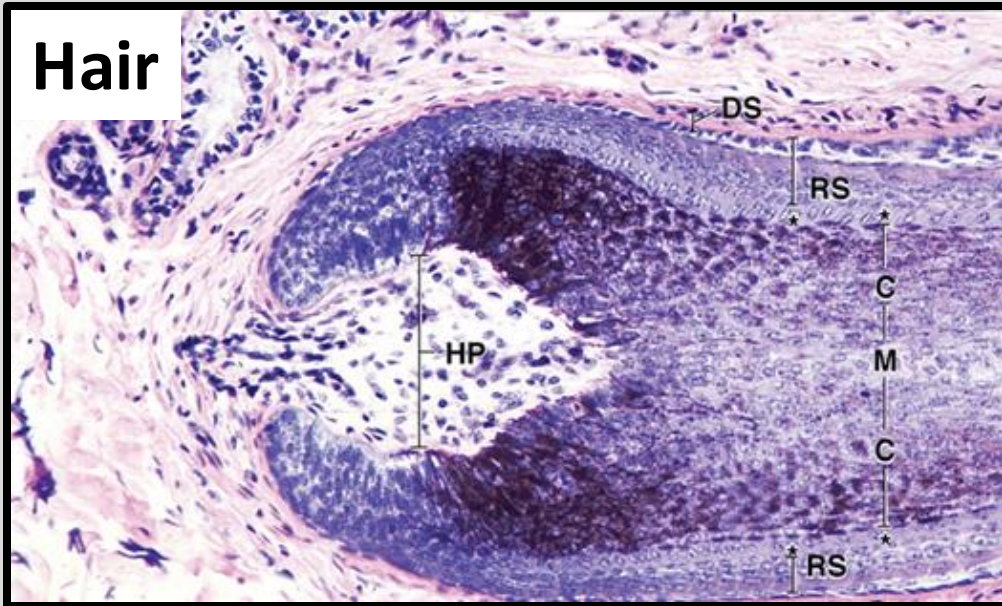


Matrix of Nails

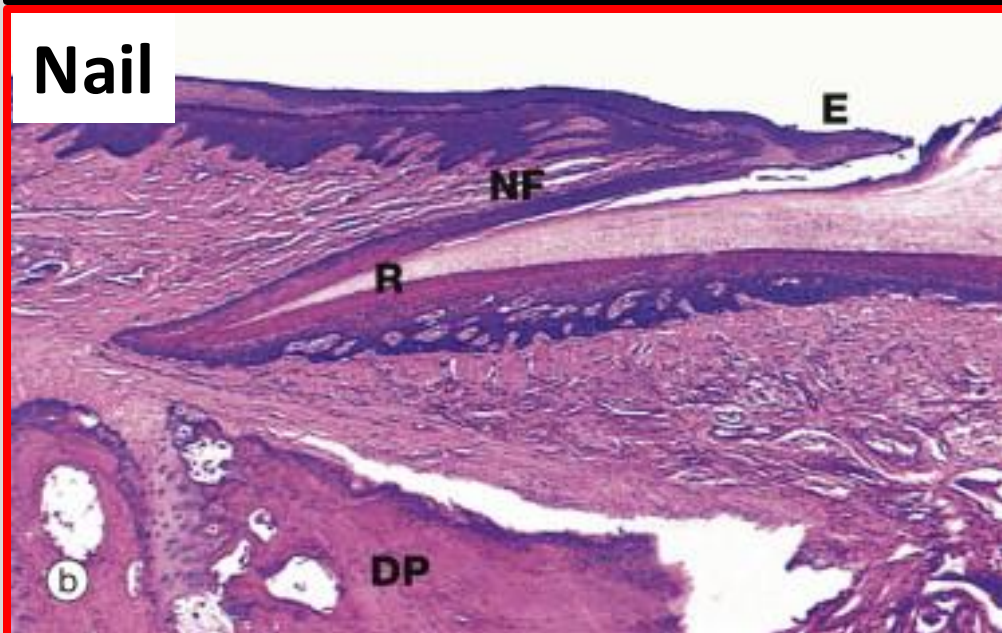
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Hair



Nail

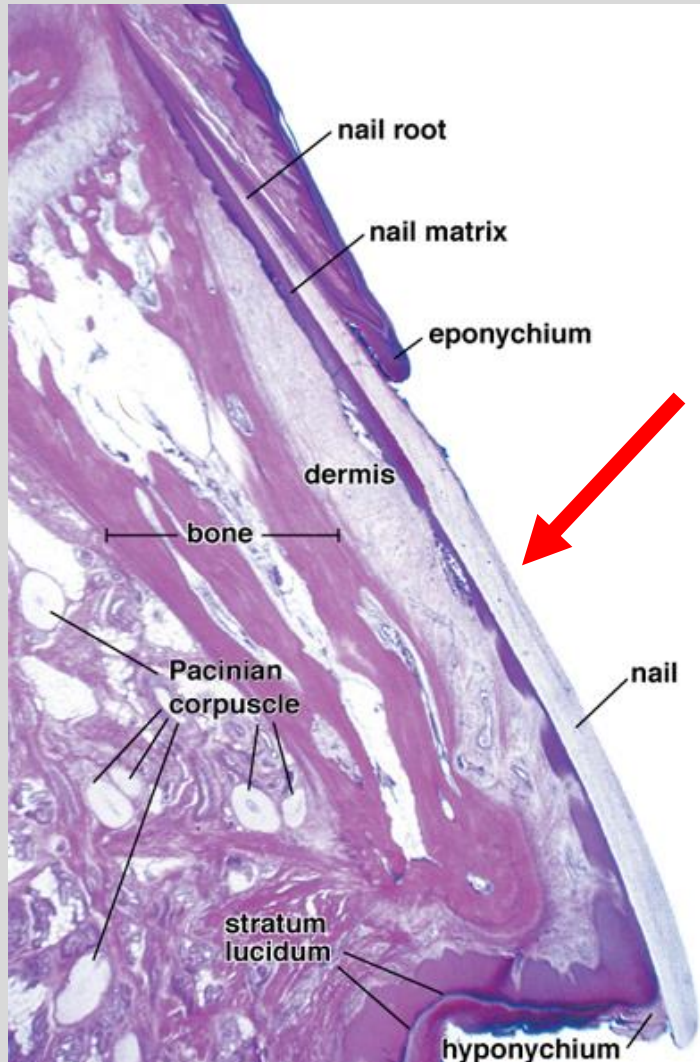


Matrix of nails has stem cells, melanocytes, Merkel's and Langerhans' cells

The stem cells divide, migrate toward the root and differentiate and produce keratin of the nail which is hard keratin, does not desquamate – consists of densely packed keratin filaments embedded in a matrix of amorphous keratin with a high sulfur content, which makes nails hard.

Crescent-shaped white area near nail root is the lunula: color is from thick, opaque layer of partially keratinized matrix cells

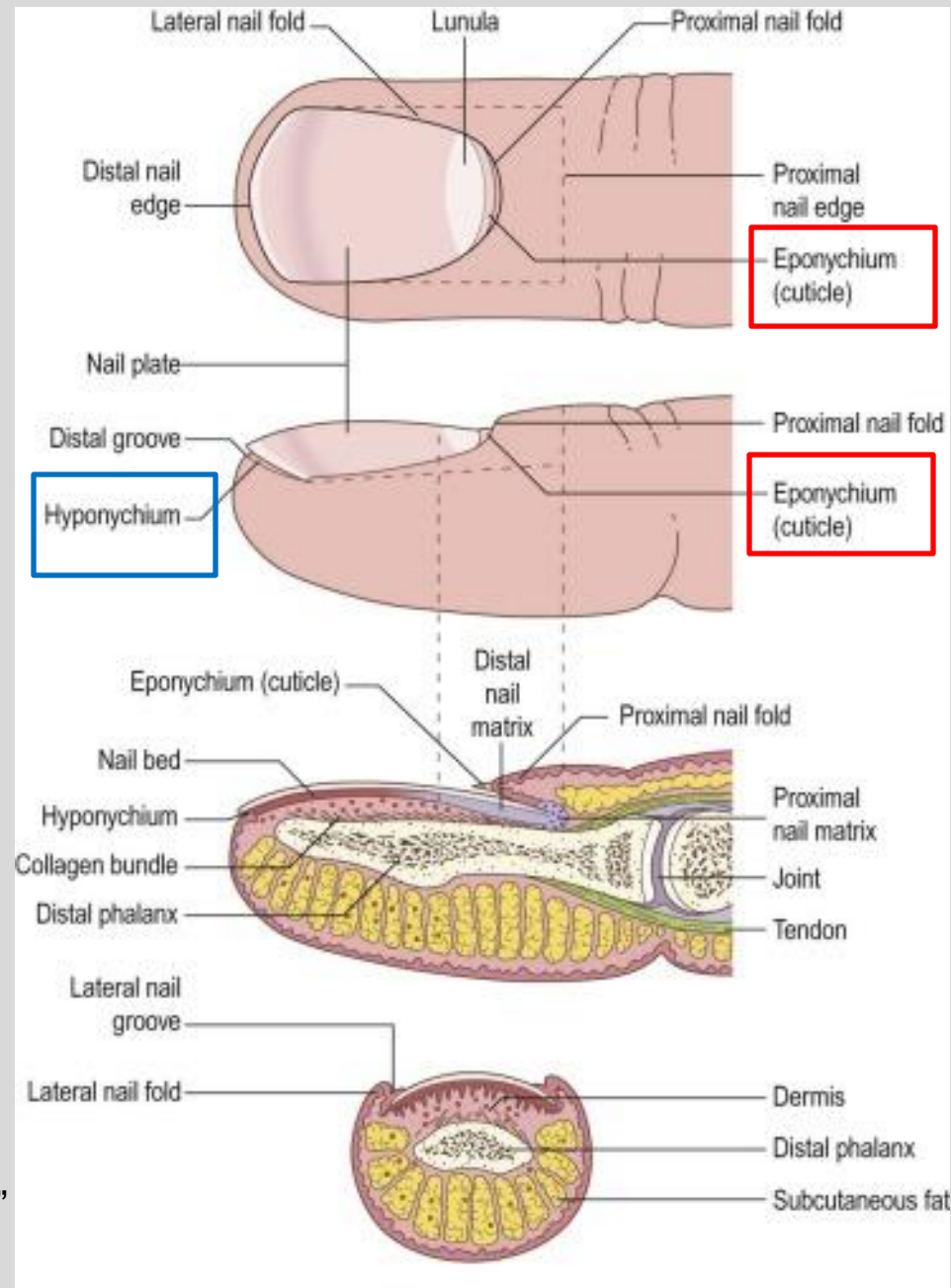
Nails are a proxy for the digits' stratum corneum



Nail bed consists of epithelial cells continuous with stratum basale and spinosum of epidermis; there is no stratum granulosum and the nail itself serves as the stratum corneum



For the nail,
but not of
the nail



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Eponychium (cuticle) – edge of skin fold covering nail root, also hard keratin, does not desquamate

Hyponychium – epidermis underlying distal tip of nail where stratum corneum of distal free edge contacts the nail; thickened epidermal layer which secures the free edge of the nail plate at fingertip

Source: "Skin and its appendages"
in Gray's Anatomy (2010)

Dark coloration of nails, melanonychia



Increased melanin expression
This example is “pseudo-Hutchinson’s sign”.
Benign.

Source: Theunuo et al (2019) Pseudo-Hutchinson’s sign – Unraveled by dermoscopy



Increased melanocytes
This example is “Hutchinson’s sign”.
Acral lentiginous melanoma, serious cancer.

Source: <https://my.clevelandclinic.org/health/diseases/subungual-melanoma>

Planting the seeds (sebaceous glands)



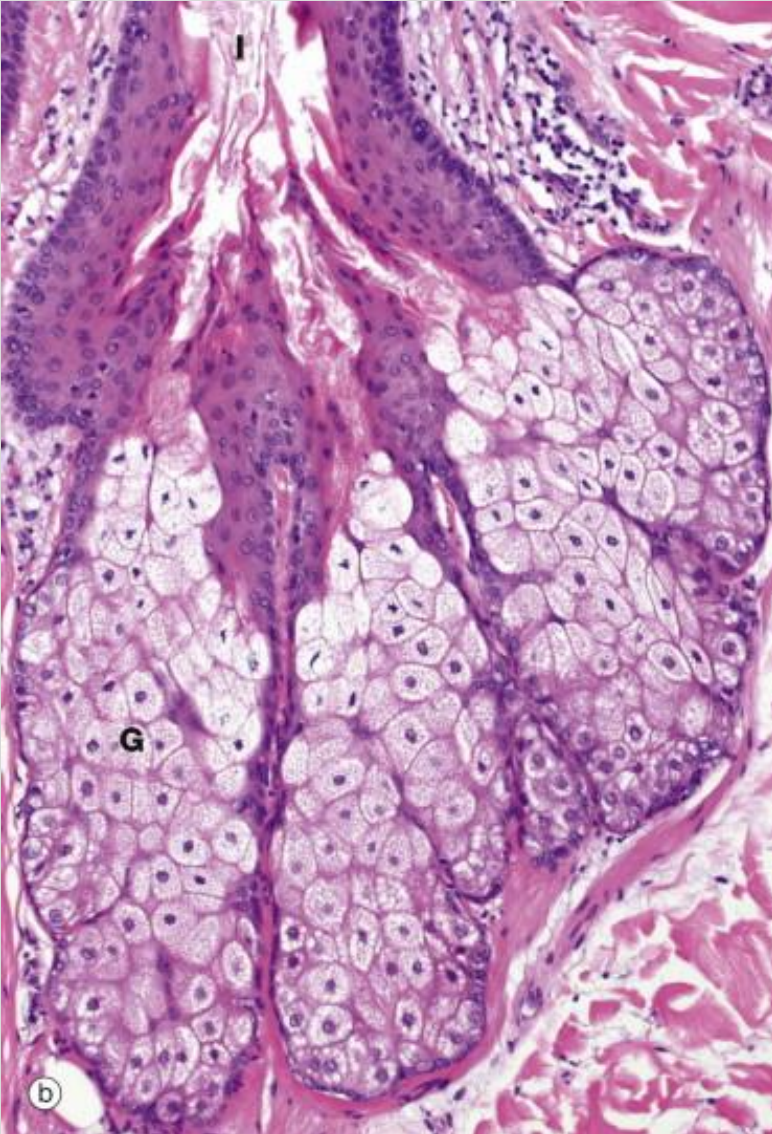
Cells of external root sheath in ~ upper 1/3 of follicle grow out into adjacent dermis and differentiate into sebaceous glands

Sebaceous cells develop a product that kills the cell.

See a theme?



Planting the seeds (sebaceous glands)



Sebaceous glands are holocrine glands.
Whole cell dies in excretory process.

Secrete sebum - fatty material that oils and
lubricates the hairs.
May possess bacterial and fungicidal
properties.

Ducts open up into follicle at the outgrowth,
several sebaceous glands per follicle.



Sebaceous hyperplasia

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© Jere Mammino, DO

Punctate swelling of the epidermis which reflect abnormal # sebaceous glands at that follicle.

~1% of population, typically middle-aged men.

Benign.

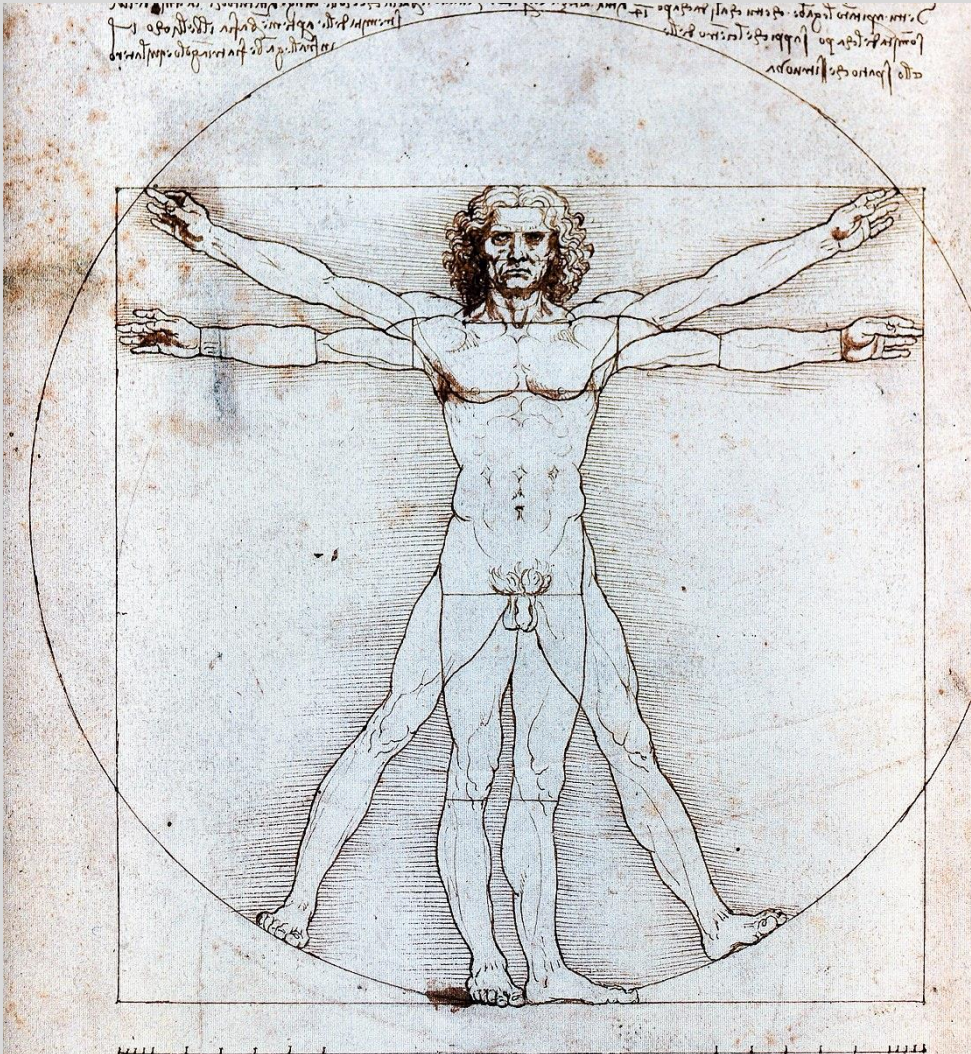
Looks like basal cell carcinoma.



Sweat glands

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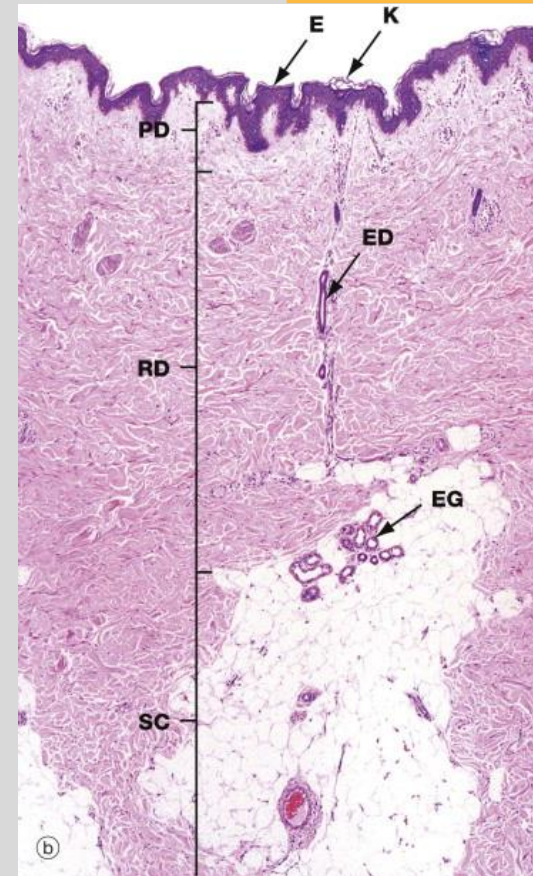
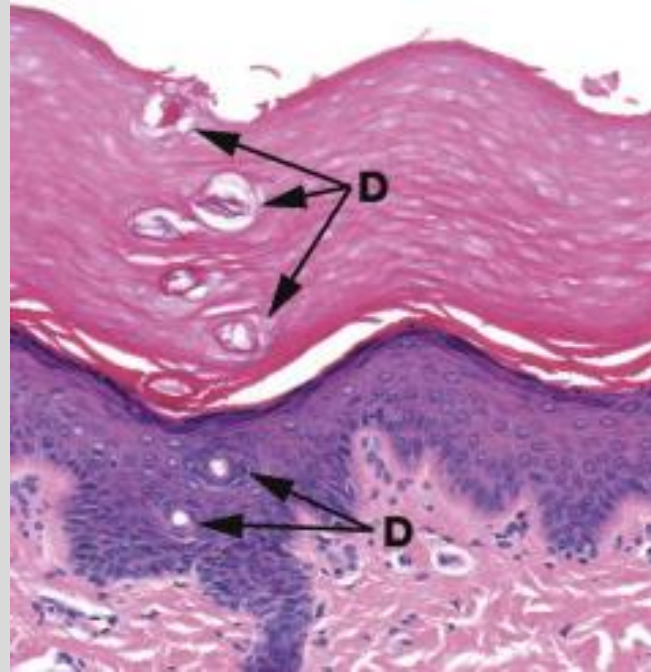
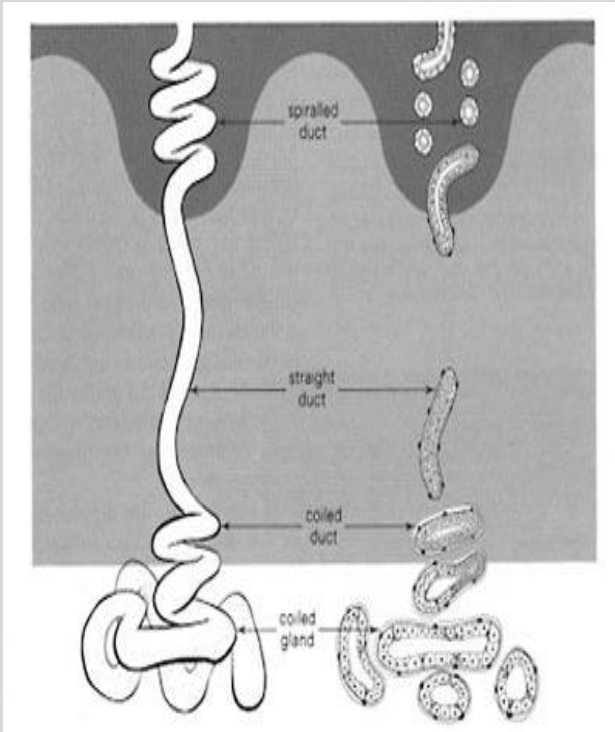


Eccrine – most common, distributed all over body* except lips and part of the external genitalia

Apocrine – limited to axilla, areola, nipple, skin around anus and external genitalia; also, ceruminous glands of external meatus of the ear and glands of eyelashes.

**We could be named homo sudore (sweat man) rather than homo sapiens (wise man) since we are the only creature with eccrine glands distributed throughout the body!*

Eccrine gland



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Simple tubular gland with a secretory portion and excretory duct. Secretory part housed in deep dermis or subcutaneous tissue. Excretory duct goes directly to skin surface.

Reabsorption of some Na and water in duct results in release of hypotonic sweat low in protein with different amounts of NaCl, urea, uric acid, and ammonia



Major role in temperature regulation by evaporation of water from sweat on body surface

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

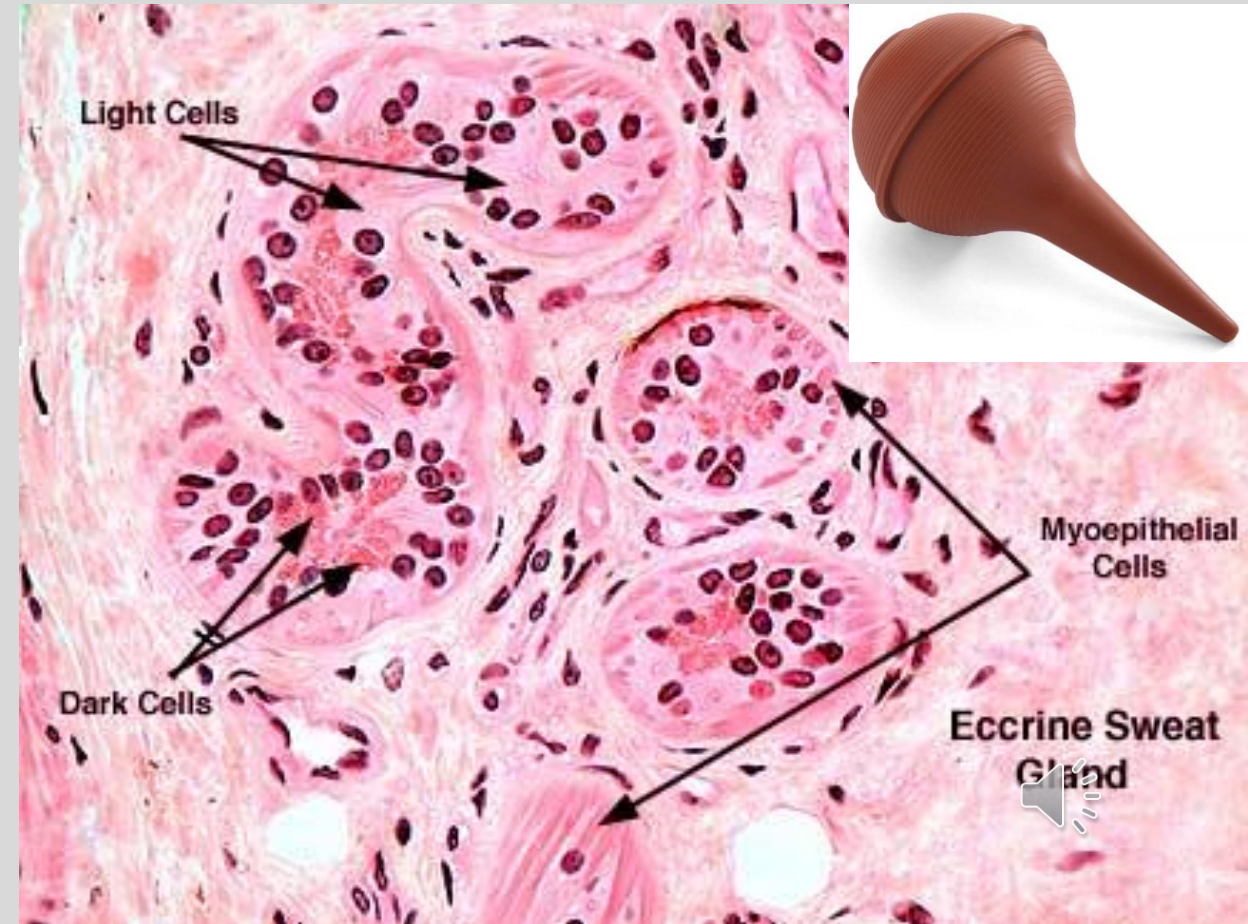
Cells of the eccrine gland

Secretion is similar to ultrafiltrate of blood.
No loss of cytoplasm (merocrine secretion, exocytosis)

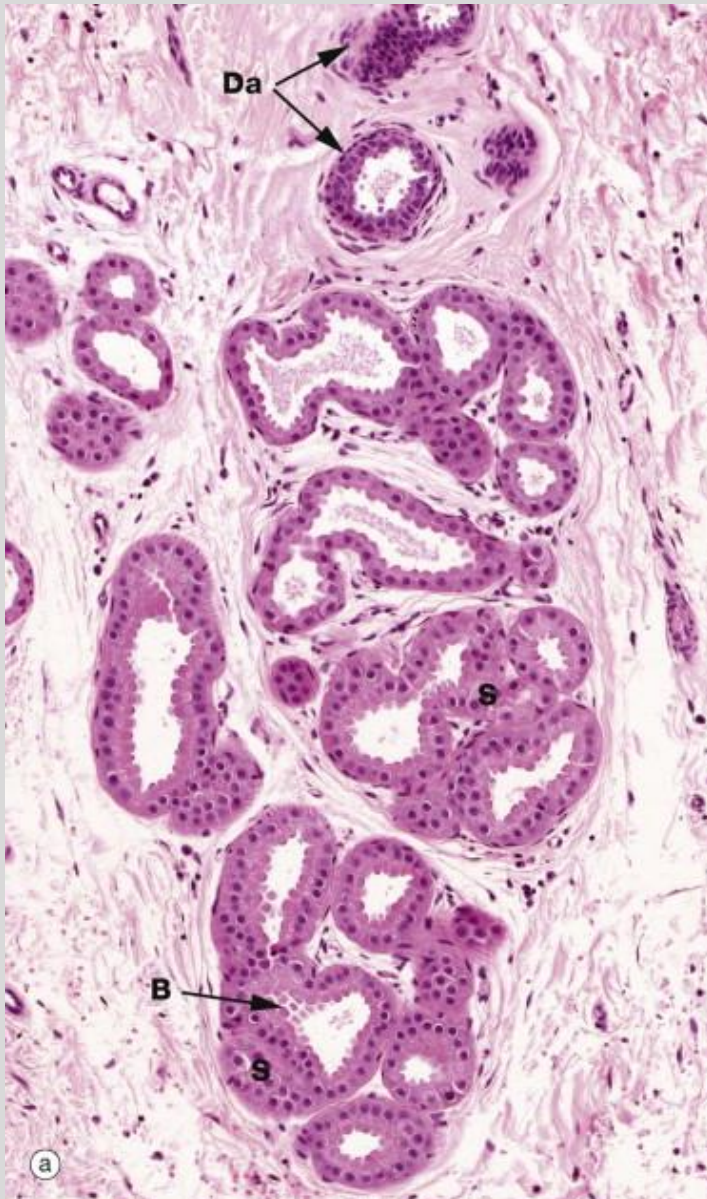
Clear cells – abundant glycogen, pale cytoplasm, cuboidal to simple columnar; secrete H₂O, NaCl, & urea. Canaliculi conduct sweat to lumen of ducts

Dark cells – secretory granules, secrete glycoprotein (proteoglycans)

Myoepithelial cells – oriented obliquely and longitudinally around secretory portions of tubules, contractions aid in expelling sweat; the excretory duct does NOT have myoepithelial cells



Apocrine glands



Develop from epidermal downgrowths for hair follicles;
connection to follicle remains
Coiled with secretory portion deep in dermis or hypodermis

Lumens of secretory portion are much wider than those of
eccrine glands

Apocrine glands store secretory products in their lumens,
eccrine glands do not.

Also, reabsorption does not take place in duct, unlike eccrine
glands

Source: "Integumentary System" in Ross and Pawlina Histology, 7th Ed. (2015)

Cells of the apocrine glands

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Originally thought to be apocrine secretion (membrane budding)
– actually merocrine.

Only one secretory cell type – eosinophilic with bleblike apical protrusions

Myoepithelial cells are also present in secretory portion (same as eccrine)

Produce secretion with protein, carbohydrate, ammonia, lipid, and some organic compounds that may give milky color
Secretion develops odor via bacterial action on skin,

Become functional at puberty – linked to pheromones?

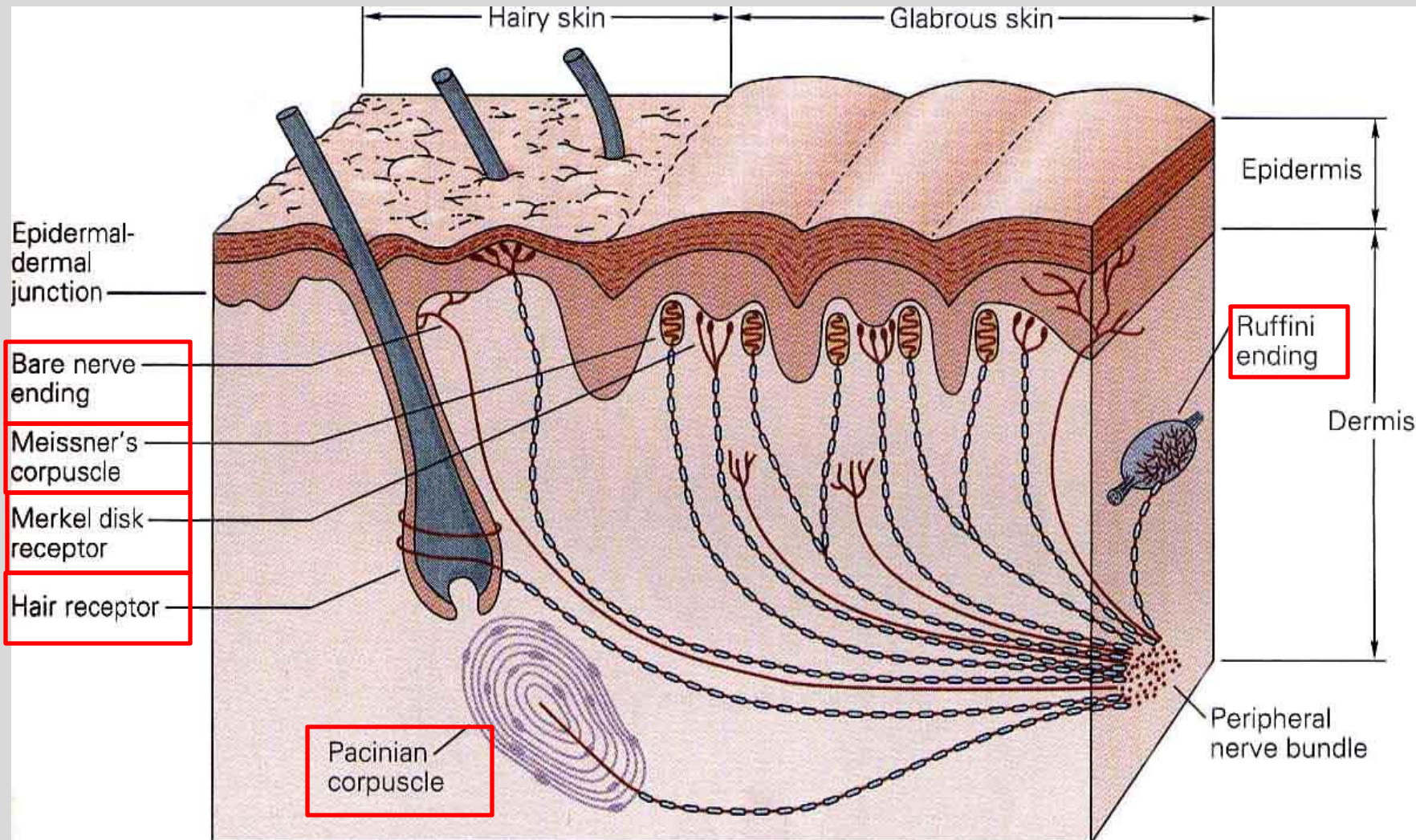
Source: “Integumentary System” in Ross and Pawlina Histology, 7th Ed. (2015)



Nerve supply of skin

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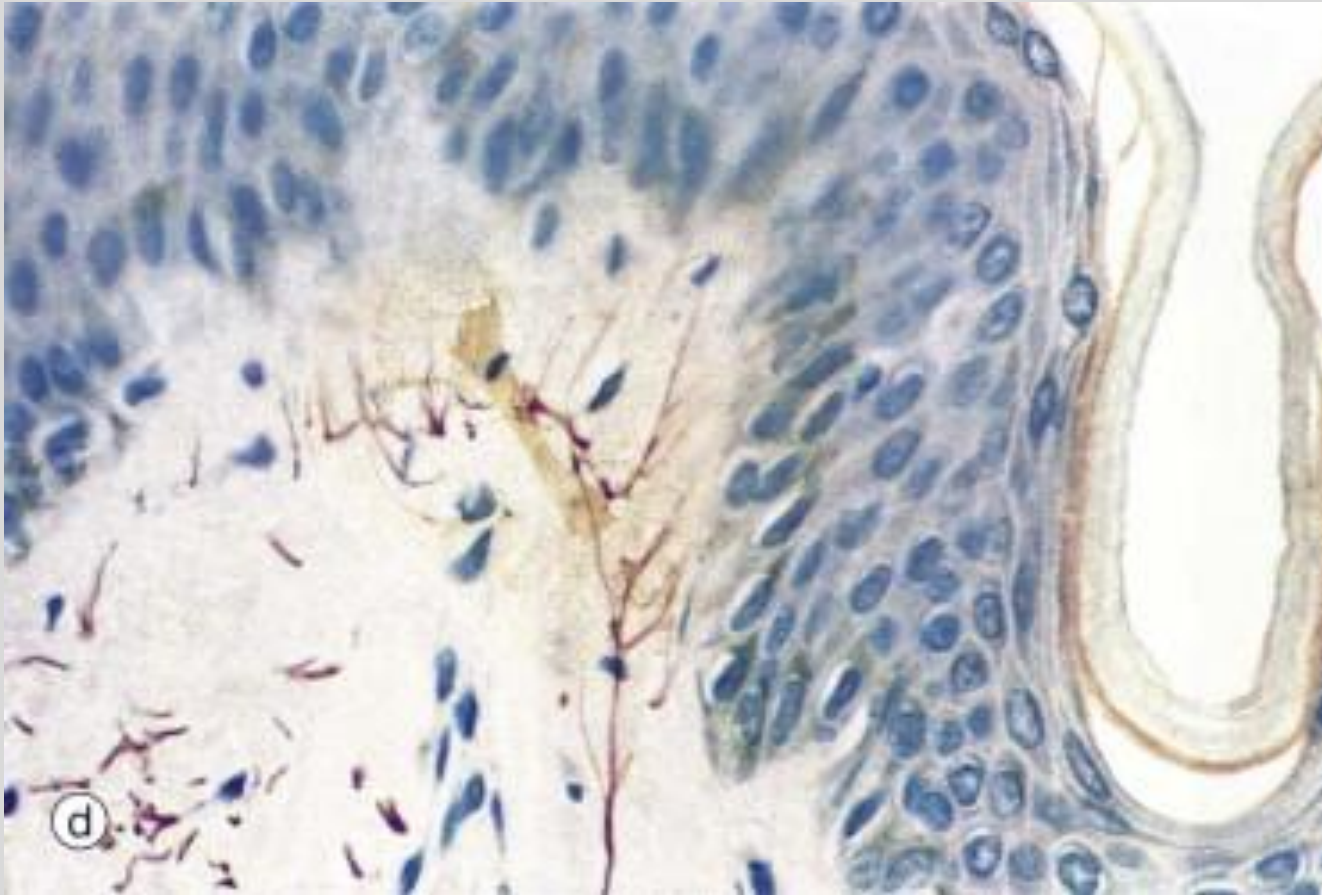


Source: Kandel and Schwartz,
Principles of Neuroscience

Free nerve endings

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Present in both throughout epidermis and dermis

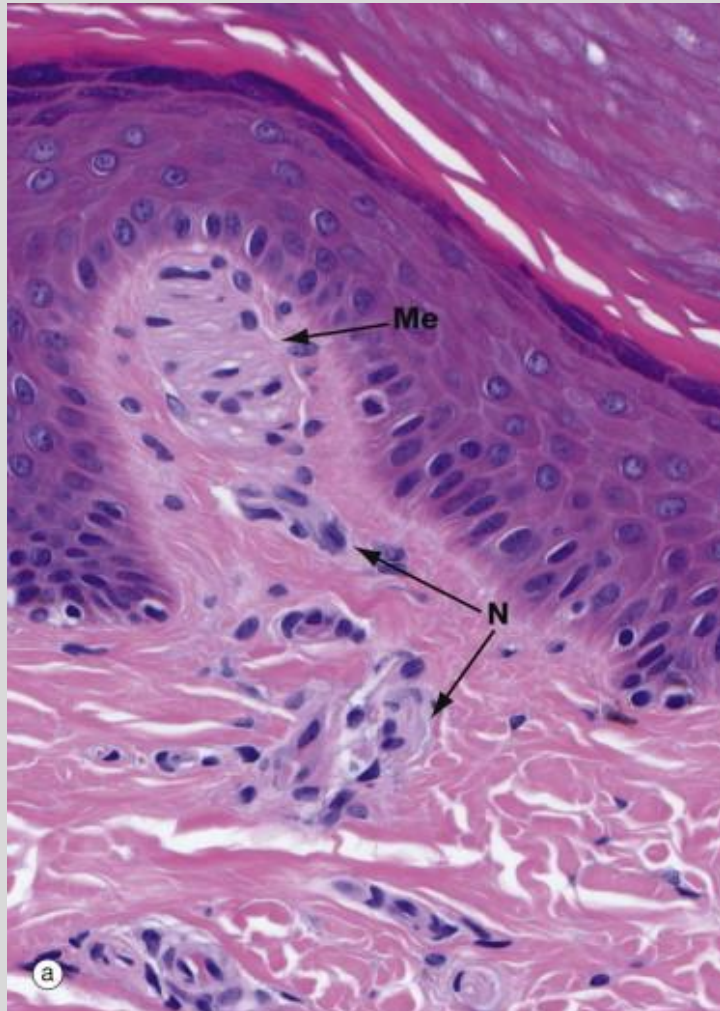
Lack a connective tissue or Schwann cell investment, i.e. “free”

Heat, cold, itch, and pain

Surround most hair follicles
mechanoreceptors (e.g. “whiskers”)



Meissner corpuscles

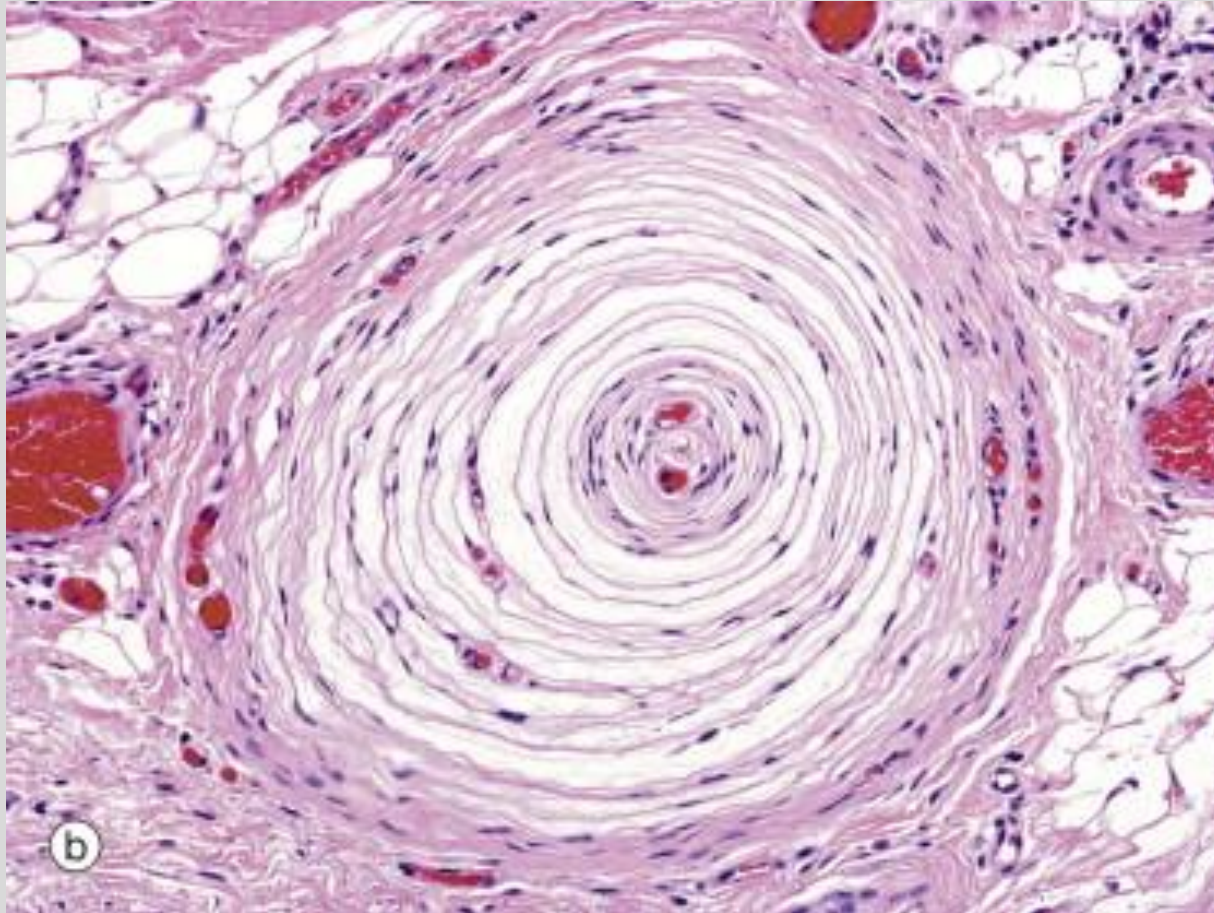


Touch receptors within dermal papillae
Especially responsive to stimuli in
papillary layer of hairless skin (lips,
palmar surfaces, etc.)
Appear like a twisted string on microscopy

Source: "Nervous Tissue" in
Wheater's Functional Histology,
6th Ed. (2014)



Pacinian corpuscles



Myelinated nerve ending surrounded by capsule, “onion” appearance

Found in deeper dermis, hypodermis, and associated with various connective tissues (e.g. ligaments and fascia)

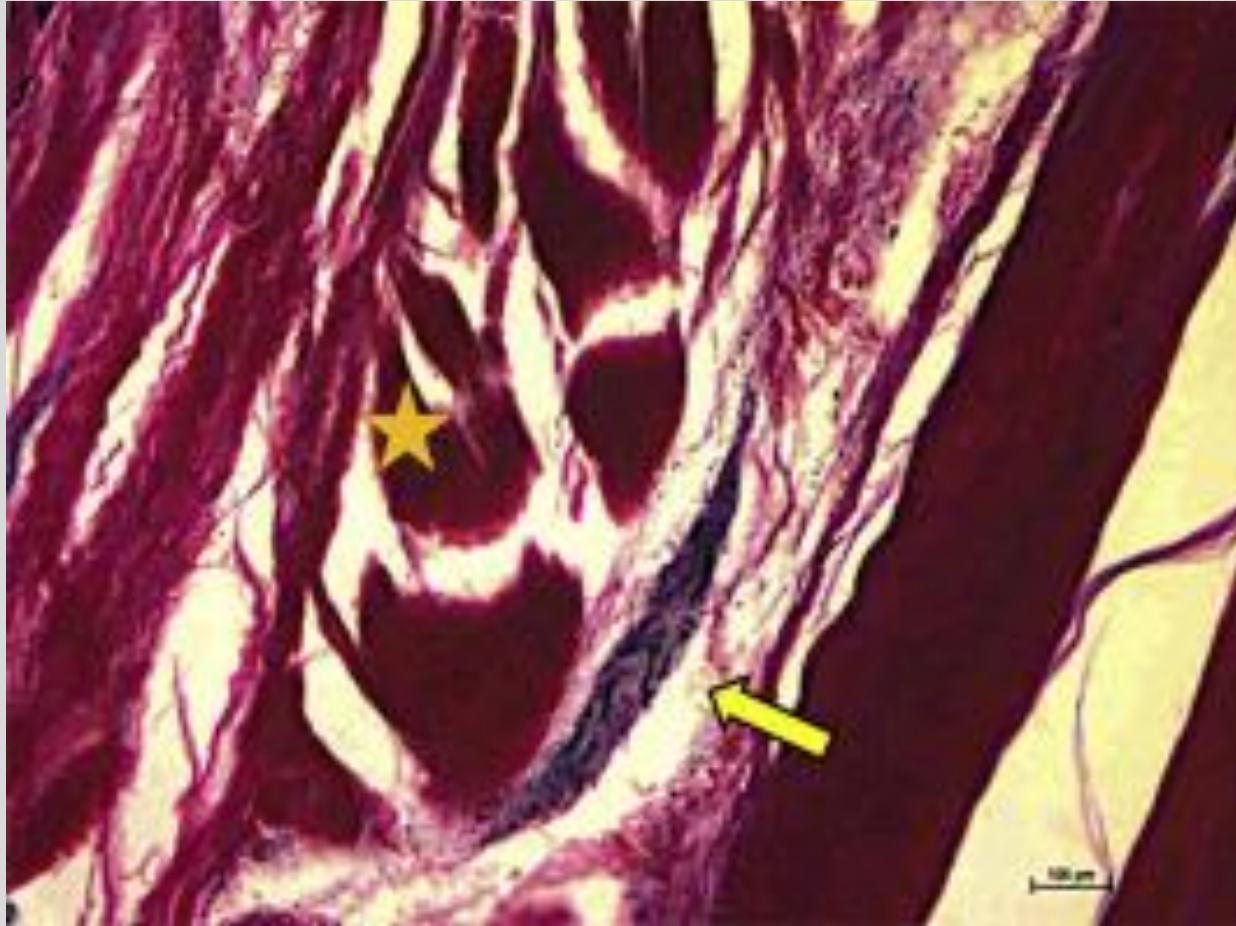
Deep pressure receptors for mechanical and vibratory pressure; enclosed in connective tissue capsule



Ruffini corpuscles

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Relatively simple “bag”

Found in dermis; also associated
with various connective tissues
(e.g. ligaments and fascia)

Detects stretch

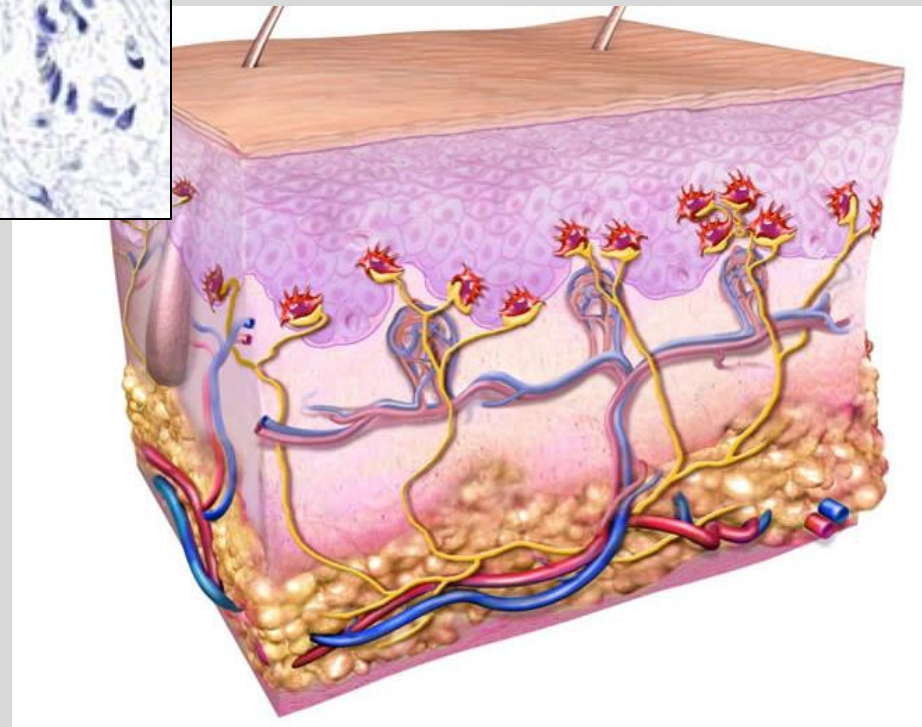
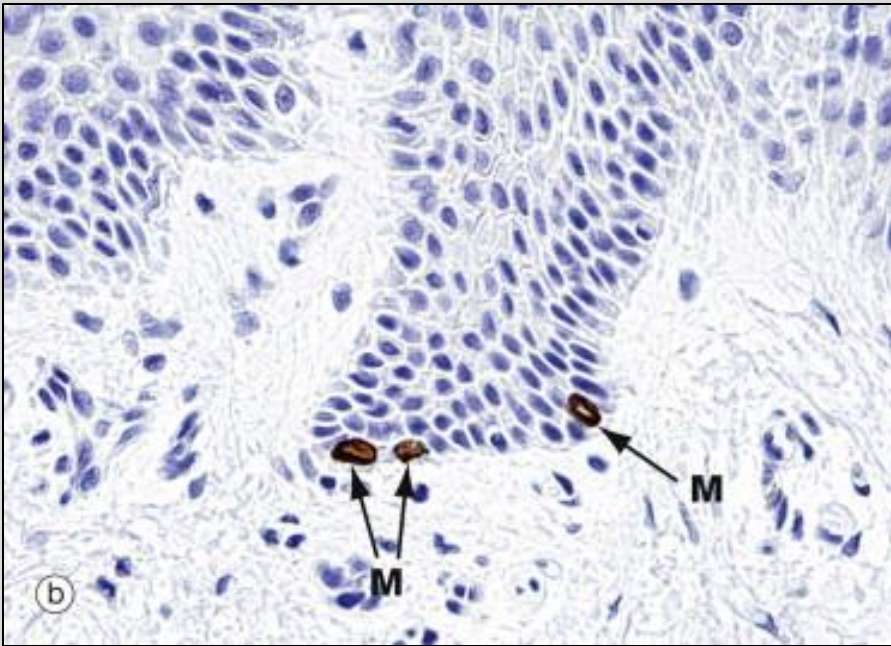


Source: Kholinne et al (2018) Mechanoreceptors distribution in the human medial collateral ligament of the elbow

Merkel cells

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Modified epidermal cells in
stratum basale

Sensory function for fine-
touch, most populous in
fingertips

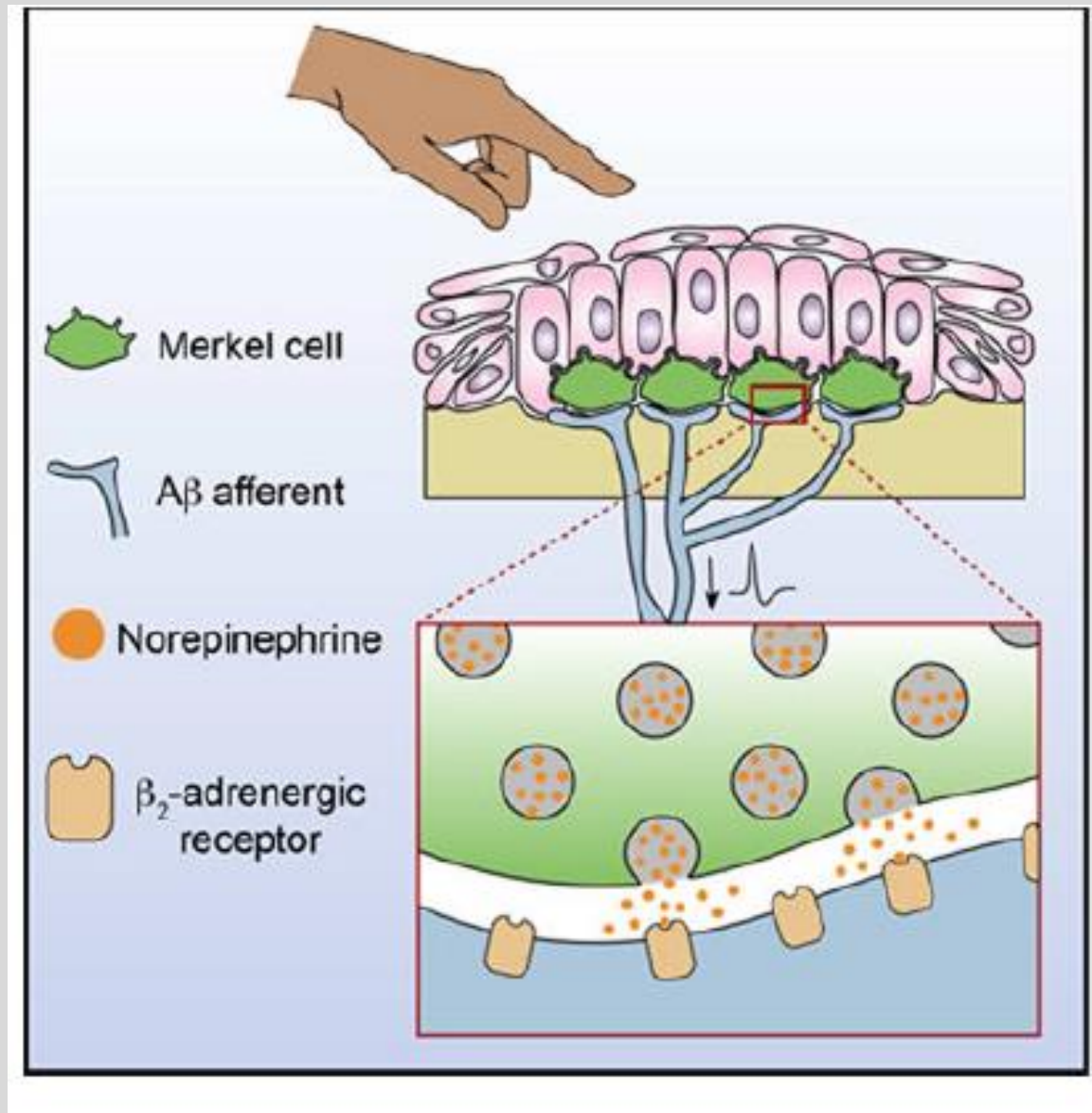
Bound to adjoining
keratinocytes by
desmosomes



Merkel cells

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Not neurons.
Derived from neural crest.
Release NE.



Summary

- Epidermal appendages include hair, nails, sebaceous glands, and sweat glands.
- Develop from epidermal downgrowth and are housed deep in skin.
- Hair, nails, and sebaceous glands express similar developmental sequence as epidermis.
- Eccrine and apocrine sweat glands differ in many ways.
- The skin has an extensive nerve supply.



Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

